Lab 3: Component Modelling & Architectural Pattern Selection

System: Self-Service Coffee Kiosk

Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Reason 1 — Separation of concerns

The touchscreen interface, order processing, payment handling, receipt printing, and menu/pricing data are separated into distinct components. This makes the system easier to understand, test, maintain, and modify.

Reason 2 — Suitable complexity

The kiosk supports a small, fixed workflow: selecting one of three coffee types and one of two sizes, paying by credit card, and printing a receipt. A layered design provides clear modularity without the deployment and monitoring overhead of microservices.

Security advantage

The User Interface does not directly access stored menu data or process payment details. The Order Manager validates order information before requesting payment, and the Payment Service should delegate card handling to a compliant payment terminal/provider. Card details should not be stored in the menu database.

Performance benefit

Separating presentation, order processing, and data access lets each component perform a focused task. Menu/pricing queries are handled by the data component, while the Order Manager coordinates the order and payment workflow without mixing it with touchscreen rendering.